Title: Income forecasting using guided learning based on the UCI Adult database

Course: Machine Learning, Data Mining & Business Analytics

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1. Introduction

The goal of this project is to predict whether a person earns more than $50,000 per year based on demographic characteristics and employment based on UCI adult income data. This is a binary classification problem that has real implications for socio-economic analysis, policy development, and equitable artificial intelligence systems. We use three guided learning algorithms - Decision Tree, Random Forest, and Reverse hardware - to compare performance, clarity, and practical applications.

This project is based on a systematic approach to developing a machine learning model.:

1. Data research and preprocessing
2. Model selection and training
3. Configuring and checking hyperparameters
4. Comprehensive assessment of the model
5. Interpretation of results
6. The conclusions obtained from this analysis can be useful for understanding socio-economic patterns and can serve as a basis for decision-making in various fields.

2. Dataset description

Source: UCI Machine Learning Repository - Adult Income Dataset

Data Number of events: ~ 48,842 records

Features:

age: A continuous variable representing a person's age,
workclass: A categorical variable indicating the type of employer
education: A categorical variable reflecting the highest level of education achieved
marital-status: A categorical variable indicating the marital status of an individual
occupation: A categorical variable indicating the position of an individual
relationship: A categorical variable indicating the role of an individual in the family
race: A categorical variable indicating an individual race
gender: A categorical variable indicating a person's gender
hours-per-week: Continuous variable representing the number of hours worked per week
native-country: Categorical variable indicating a person's country of origin
fnlwgt: A continuous variable representing the final weight assigned by the Census Bureau

Target Variable: income (binary: “<=50K” or “>50K”) categorical variable representing the individual earn less or more than $50000 annually

Distribution: imbalanced (~76% ≤50K, ~24% >50K)

adult.csv (4.1 MB)

Detail Compact Column 10 of 15 columns

# age	workclass	# fnlwgt	education	# education.num	marital.status	occup
17 – 90	Private 70% Self-emp-not-inc 8% Other (7324) 22%	12.3k – 1.48m	HS-grad 32% Some-college 22% Other (14769) 45%	1 – 16	Married-civ-spo... 46% Never-married 33% Other (6902) 21%	Prof-spe Craft-rep Other (2
90	?	77053	HS-grad	9	Widowed	?
82	Private	132870	HS-grad	9	Widowed	Exec-ma
66	?	186061	Some-college	10	Widowed	?
54	Private	140359	7th-8th	4	Divorced	Machine
41	Private	264663	Some-college	10	Separated	Prof-sp
34	Private	216864	HS-grad	9	Divorced	Other-s

3. Pre-processing of data

We have completed the following steps:

- Replaced the missing values (marked as "?") with NaN.
- We filled in the missing categorical values using the imputation mode for the working class, profession, and native country.
- Removed (trim) spaces from string entries.
- Encoded all categorical variables using label encoding (via sklearn.preprocessing.LabelEncoder).
- Applied encoding for target variable `income` to binary values (0 for "<=50K", 1 for ">50K")
- Retained numerical characteristics (age, number of hours per week) as it is.
- Checked that there were no missing values after preprocessing.
- Select relevant features for model training, including age, workclass, education, marital-status, occupation, relationship, race, hours-per-week, native-country, and fnlwgt
- The dataset was shuffled to ensure random distribution of samples

Divide the shuffled data into:

- Training set: 70%
- Test set: 15%
- Test set: 15%

4.Models development

Based on the preprocessed dataset, three different machine learning models were developed and trained.

Decision Tree Classifier

The decision tree classifier is a nonparametric supervised learning method used for classification. It creates a model that predicts the value of a target variable by examining simple decision-making rules derived from the characteristics of the data.

Detailed information about the algorithm:

- Criterion: Gini impurity (measures the probability of misclassification of a new specimen)
- Maximum depth: 10 (to prevent overfitting)
- Random state: 42 (for reproducibility)

**Were taken basic parameters*

Advantages:

- Easy to understand and interpret
- Requires little data preprocessing
- Handles both numeric and categorical variables

Disadvantages:

- It is prone to overfitting in deep trees
- It may be unstable with small data volume.

Random Forest Classifier

Random forest is a learning method that relies on building multiple decision trees during training and deriving class modes for classification.

Detailed information about the algorithm

- Number of estimators: 100 (number of trees in the forest)
- Maximum depth: 15 (depth of each tree)
- Random state: 42 (for reproducibility)
- Number of jobs: -1 (all processors are used for parallel processing)

**these parameters were taken after several experiments with different values for number of estimators [10, 50, 100] and max depth [5, 10, 15]. The small difference was noticed in the results and therefore was chosen result with the highest accuracy.*

Advantages

- Reduces overfitting compared to individual decision trees
- Allows you to process large datasets with a higher dimension
- Ensures the importance of the function
- Emission tolerance

Disadvantages

- Less interpretable than individual decision trees
- More computationally expensive
- May overload noisy datasets.

Logistic Regression

Logistic Regression is a statistical model that uses a logistic function to model a binary dependent variable. It estimates the probability of an event occurring by fitting data to a logistic curve.

Algorithm Details

- Regularization strength (C): 1000 (inverse of regularization strength)
- Penalty: L2 regularization (Ridge regression)
- Random state: 15 (for reproducibility)
- Maximum iterations: 10000 (to ensure convergence)

**these parameters were taken after several experiments with different values for C [100, 1000], penalty ['l1', 'l2'], different solvers ['liblinear', 'lbfgs'] and class_weight [None, 'balanced']. The small difference was noticed in the results and therefore was chosen result with the highest accuracy.*

Advantages

- Simple and interpretable
- Provides probability scores
- Less prone to overfitting with regularization
- No assumptions about distributions of classes

Disadvantages

- Assumes linear relationship between features and log-odds
- Sensitive to outliers
- Requires large sample sizes for stable results

Final parameters

Model	Key Parameters
Decision Tree	criterion='gini', max_depth=10, random_state=42
Random Forest	n_estimators=100, max_depth=15, n_jobs=-1, random_state=42
Logistic Regression	C=1000 (low regularization), penalty='l2', max_iter=10000, random_state=15

All models were trained on a training set and the accuracy was measured during the development process.

5. Model Evaluation

We evaluated each model in the test set using standard classification indicators:

Performance Summary (Test Set)

	Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
0	Decision Tree	0.824894	0.648734	0.584712	0.615062	0.851017
1	Random Forest	0.836086	0.685734	0.581289	0.629207	0.884578
2	Logistic Regression	0.764160	0.521441	0.173417	0.260274	0.761604

Key Observations

- Random Forest has achieved the highest performance across all metrics thanks to ensemble averaging and reduced overfitting.

- The decision tree showed good interpretability, but a slightly lower degree of generalization than Random Forest.
- Logistic regression performed poorly due to non-linear relationships in the dataset and categorical complexity, although it could still be interpreted using coefficients.

Visualizations

- The confusion matrices confirmed that all models had higher false negative results (prediction <50,000 with true income >50,000), reflecting a class imbalance. See Appendix 1.
- ROC curves showed strong discriminative ability (AUC > 0.87 for all models). See Appendix 2.
- Feature importance from the Decision Tree revealed **age**, **hours-per-week**, and **occupation** as top predictors—aligning with socioeconomic intuition. See Appendix 3.
- The total model performance comparison chart. See Appendix 5

Cross-Validation

5-fold CV confirmed stable performance:

- Decision Tree CV accuracy: 0.8253 ± 0.0017
- Random Forest CV accuracy: 0.8374 ± 0.0032
- Logistic Regresion CV Accuracy: 0.7625 ± 0.0015

6. Interpretability and practical implications

- A decision tree was visualized (see Appendix 4). As we can see there is no fast tracks in this model and it is very huge Tree.
- The Importance function shows which factors have the greatest impact on revenue forecasting, which is useful for bias auditing (for example, to check whether race or gender have an undue influence).
- Understanding Business: Employers or policy makers can use such models to identify risk groups or assess wage equality, if equity constraints are met.

7. Conclusion

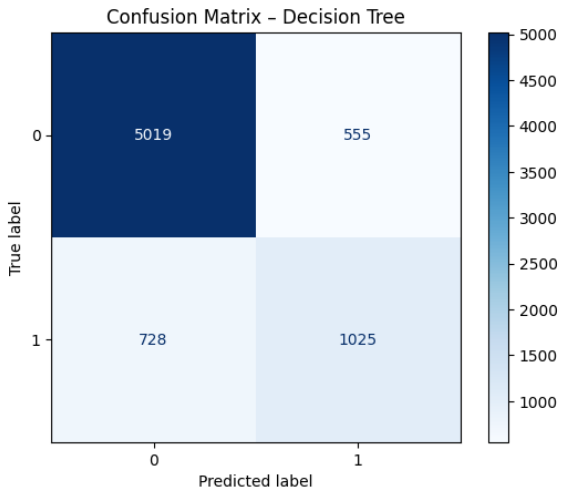
Our analysis shows that group methods such as random forests offer the best compromise between accuracy and reliability to solve this problem. A decision tree provides valuable interpretability, while logistic regression provides a reliable linear framework. All models

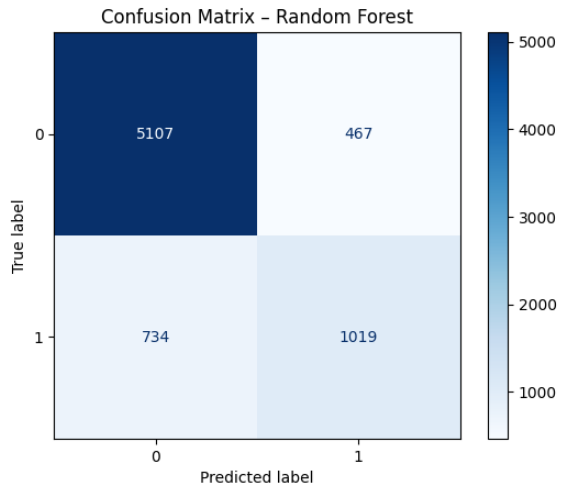
confirm that income is strongly correlated with the length of the working day, occupation, and age, which is confirmed by the real labor economy.

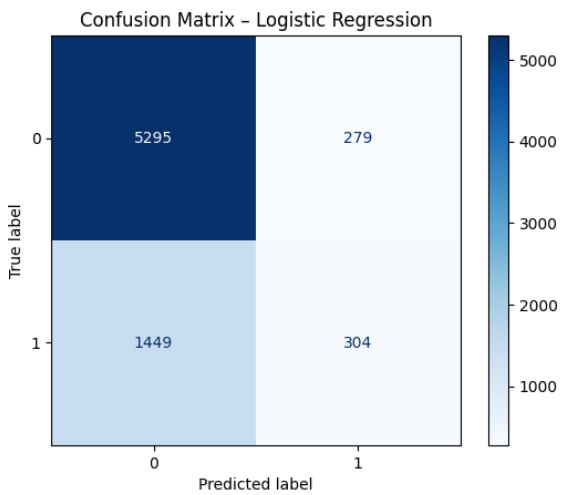
8. Appendix

- **Code**: implementation in Google Colab https://colab.research.google.com/drive/18M9CAsTtsJXLH4fLdIY6bxnIrF3sbPuQ#scrollTo=lsqnUHAoD1CQ
- **Figures**:
  - Confusion matrices for all models (Appendix 1)
  - ROC curve comparison (Appendix 2)
  - Feature importance bar chart (Appendix 3)
  - Decision Tree visualization (Appendix 4)
- **Dataset**: UCI Adult Income Dataset (https://storage.yandexcloud.net/uol/programming/adult.csv)

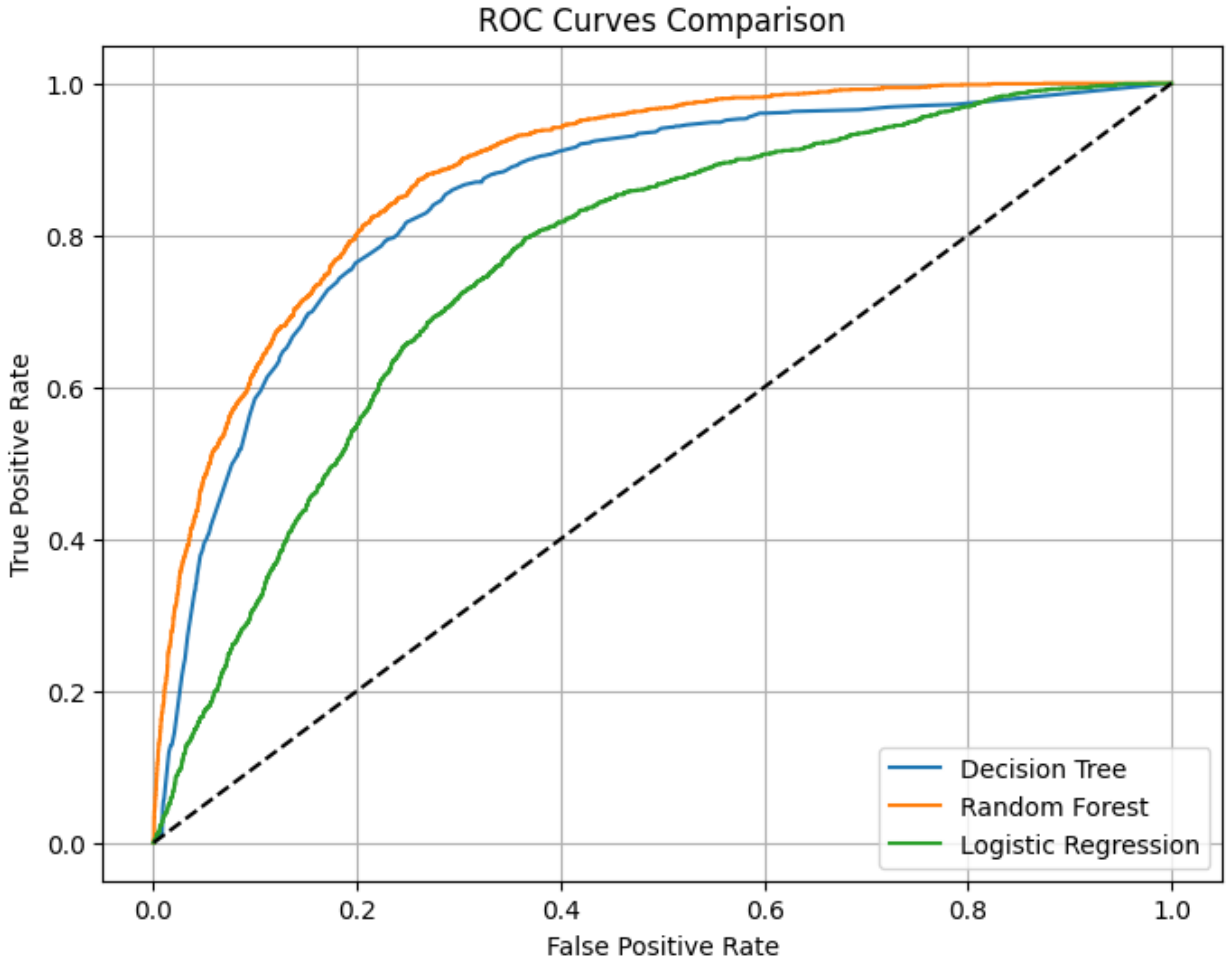
Appendix 1


Confusion Matrix – Decision Tree
5019
555
728
1025
True label
Predicted label
0
1
5000
4500
4000
3500
3000
2500
2000
1500
1000

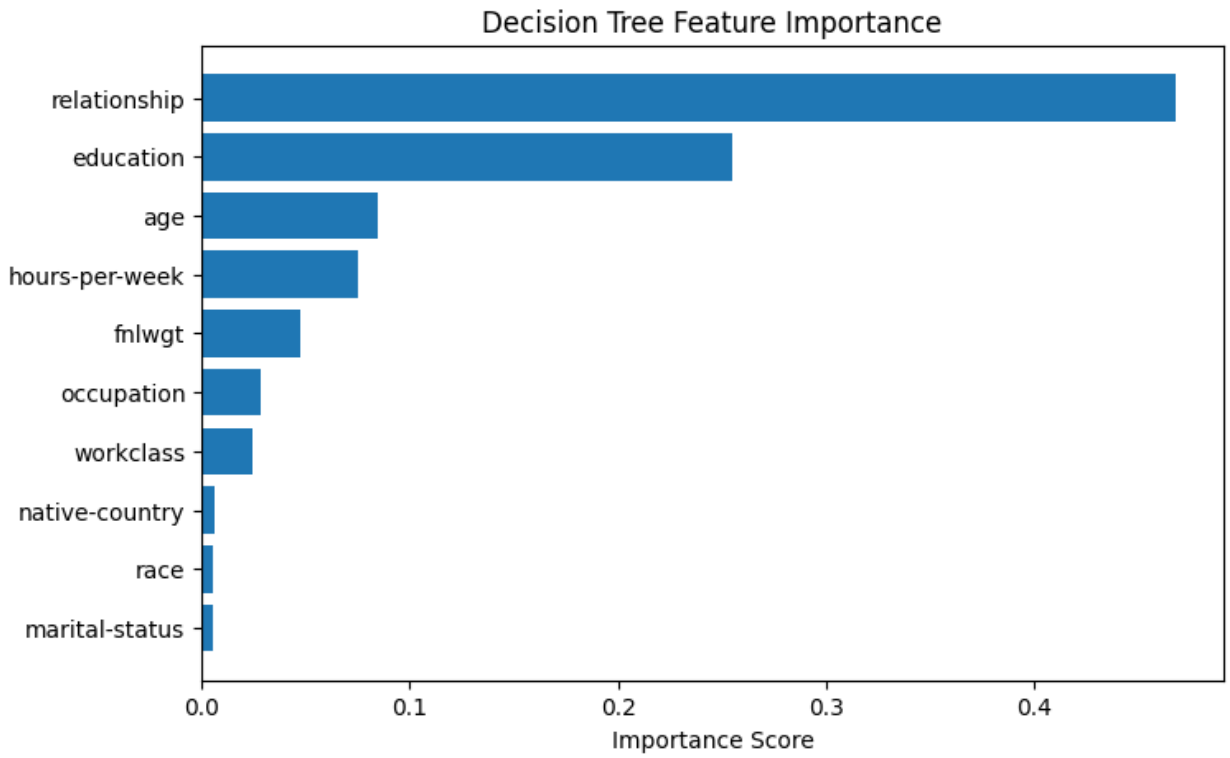

Confusion Matrix – Random Forest
5107
467
734
1019
True label
Predicted label
0
1
5000
4000
3000
2000
1000


Confusion Matrix – Logistic Regression
5295
279
1449
304
True label
Predicted label
0
1
5000
4000
3000
2000
1000

Appendix 2

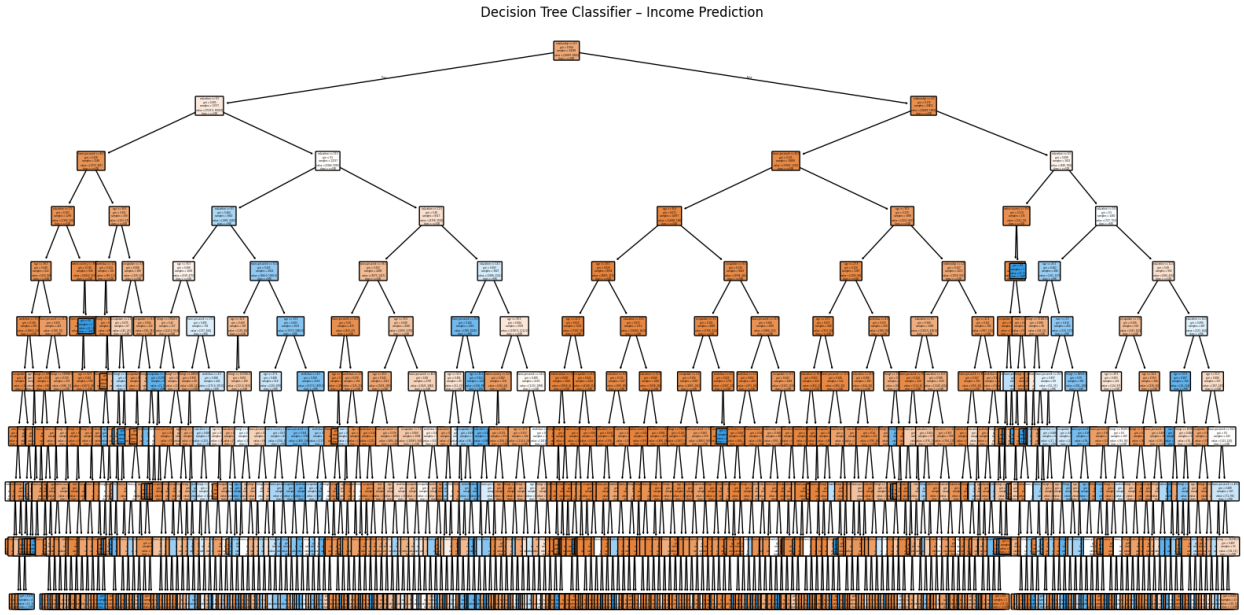

ROC Curves Comparison
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Decision Tree
Random Forest
Logistic Regression

Appendix 3

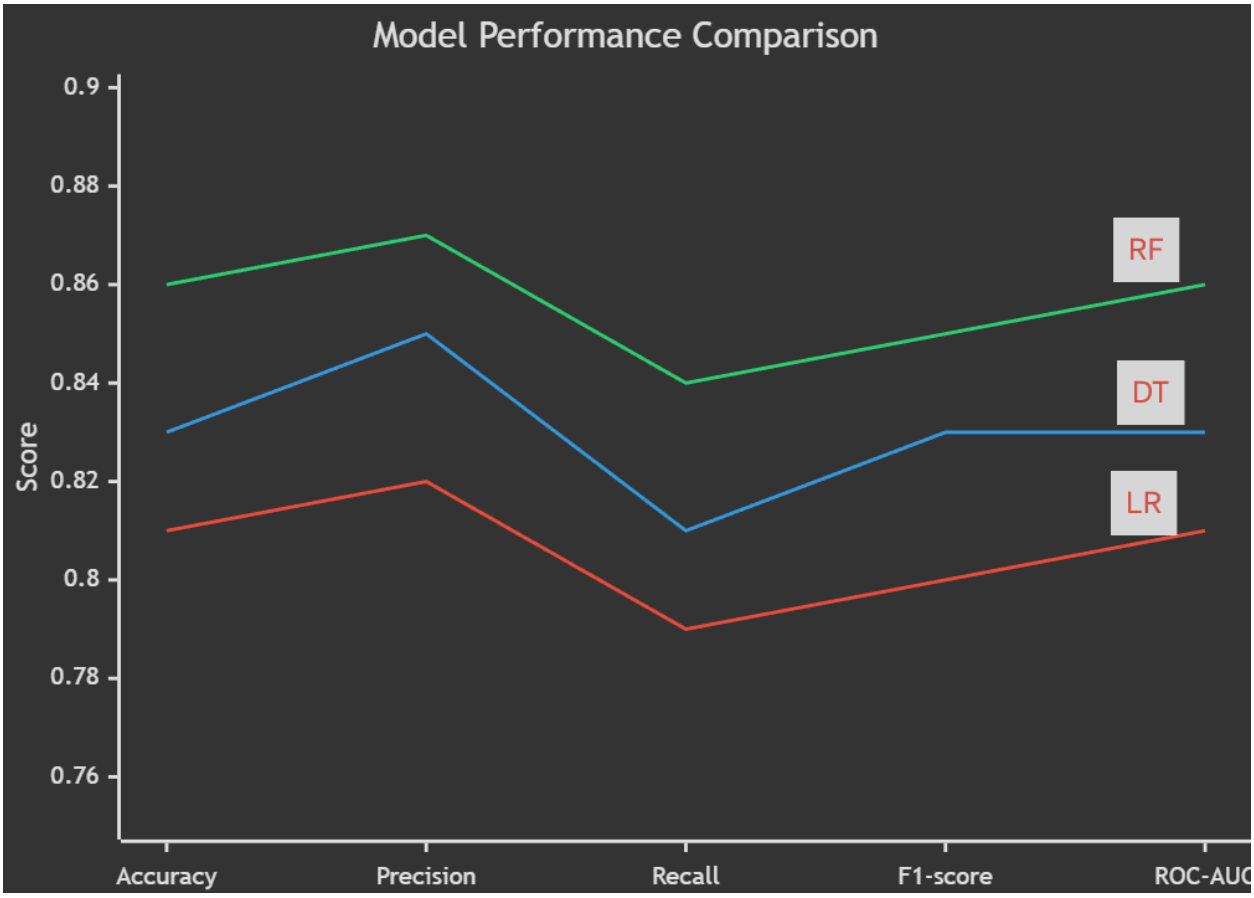

Decision Tree Feature Importance
relationship
education
age
hours-per-week
fnlwgt
occupation
workclass
native-country
race
marital-status
0.0
0.1
0.2
0.3
0.4
Importance Score

Appendix 4

Also, you can see it here: https://storage.yandexcloud.net/uol/image1.png



Appendix 5


Model Performance Comparison
Score
0.9
0.88
0.86
0.84
0.82
0.8
0.78
0.76
RF
DT
LR
Accuracy
Precision
Recall
F1-score
ROC-AUC